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EUROPEAN SEARCH REPORT

Application Number
EP 22 18 8672

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X Y	US 2018/348343 A1 (ACHOUR MAHA [US] ET AL) 6 December 2018 (2018-12-06) * paragraphs [0023], [0082], [0077]; figure 9 *	1-3,8,9, 14,15 4,5	INV. G01S7/41 G06N3/04 G06N3/08
Y	CHEIKH K ET AL: "Application of neural networks to radar signal detection in <E1>K</E1>-distributed clutter", IEE PROCEEDINGS: RADAR, SONAR & NAVIGATION, INSTITUTION OF ELECTRICAL ENGINEERS, GB, vol. 153, no. 5, 19 October 2006 (2006-10-19), pages 460-466, XP006027295, ISSN: 1350-2395, DOI: 10.1049/IP-RSN:20050103 * page 464, column 2 - page 465, column 1 *	4,5	ADD. G01S13/89 G01S13/931 G01S13/72 G01S13/34
A	US 2020/272148 A1 (KARASEV VASILIY [US] ET AL) 27 August 2020 (2020-08-27) * paragraphs [0008], [0016], [0051] *	6,7	TECHNICAL FIELDS SEARCHED (IPC) G01S G06N
A	US 2021/116540 A1 (SANTRA AVIK [DE] ET AL) 22 April 2021 (2021-04-22) * paragraphs [0043], [0067] *	6,7	

~~The present search report has been drawn up for all claims~~

Place of search	Date of completion of the search	Examiner
Munich	5 April 2023	Ferrara, Michele

CATEGORY OF CITED DOCUMENTS

X : particularly relevant if taken alone
Y : particularly relevant if combined with another document of the same category
A : technological background
O : non-written disclosure
P : intermediate document

T : theory or principle underlying the invention
E : earlier patent document, but published on, or after the filing date
D : document cited in the application
L : document cited for other reasons

& : member of the same patent family, corresponding document

EPO FORM 1503 03.82 (P04C01)

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CLAIMS INCURRING FEES

The present European patent application comprised at the time of filing claims for which payment was due.

☐ Only part of the claims have been paid within the prescribed time limit. The present European search report has been drawn up for those claims for which no payment was due and for those claims for which claims fees have been paid, namely claim(s):

☐ No claims fees have been paid within the prescribed time limit. The present European search report has been drawn up for those claims for which no payment was due.

LACK OF UNITY OF INVENTION

The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

see sheet B

☐ All further search fees have been paid within the fixed time limit. The present European search report has been drawn up for all claims.

☐ As all searchable claims could be searched without effort justifying an additional fee, the Search Division did not invite payment of any additional fee.

☒ Only part of the further search fees have been paid within the fixed time limit. The present European search report has been drawn up for those parts of the European patent application which relate to the inventions in respect of which search fees have been paid, namely claims:

6, 7

☐ None of the further search fees have been paid within the fixed time limit. The present European search report has been drawn up for those parts of the European patent application which relate to the invention first mentioned in the claims, namely claims:

☐ The present supplementary European search report has been drawn up for those parts of the European patent application which relate to the invention first mentioned in the claims (Rule 164 (1) EPC).

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LACK OF UNITY OF INVENTION
SHEET B

Application Number
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The Search Division considers that the present European patent application does not comply with the requirements of unity of invention and relates to several inventions or groups of inventions, namely:

1. claims: 1-5, 8, 9, 14, 15

   generating the first radar image by masking a range-angle image (RAI) or range-Doppler image (RDI) based on an output of an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector

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2. claims: 6, 7

   generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap

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3. claims: 10-13

   after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning

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ANNEX TO THE EUROPEAN SEARCH REPORT ON EUROPEAN PATENT APPLICATION NO. EP 22 18 8672

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report. The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

05-04-2023

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